

End-to-End Data Analytics Project: Olist E-Commerce Pipeline

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1. Executive Summary

The purpose of this project was to build a complete end-to-end data analytics pipeline for the Olist Brazilian E-Commerce dataset. The dataset contains more than 100,000 orders from 2016–2018, including information about customers, products, sellers, payments, reviews, and geolocation data.

The main objective of this project was to simulate a real-world enterprise analytics environment where data is distributed across multiple platforms and must be integrated into a centralized business intelligence system. Data was processed using PostgreSQL, Microsoft Excel, Python (Pandas), Google Sheets, and Power BI.

The project involved several stages including data ingestion, cleaning, transformation, integration, data modeling, and dashboard development.

After building the analytics pipeline and dashboard, several key insights were discovered:

Key Insights

1. **Revenue Growth Trend:** Revenue showed a steady growth trend from 2016 to 2018, indicating an increasing adoption of e-commerce services in Brazil.
2. **Delivery Performance Issues:** Some regions experienced a higher percentage of delivery delays, suggesting logistics inefficiencies in certain geographic areas.
3. **Customer Value Segmentation:** Using RFM segmentation, customers were grouped based on recency, frequency, and monetary value. This helped identify high-value loyal customers and inactive customers, enabling targeted marketing strategies.

The final dashboard enables executives to monitor sales performance, analyze logistics performance, and evaluate customer behavior in a clear and interactive format.

2. Problem Statement

Modern e-commerce platforms generate large volumes of data across multiple systems. Organizations require effective analytics pipelines to transform raw transactional data into actionable business insights.

The objective of this project was to analyze the Olist e-commerce dataset and build a reporting system that tracks key business metrics such as:

- Total Revenue
- Number of Orders
- Average Order Value
- Delivery Performance
- Customer Segmentation
- Product Category Performance

The main challenge was that the dataset was fragmented across multiple tables and required integration from different processing tools. Therefore, the project aimed to simulate a real corporate data environment where data is stored and processed across different platforms before being consolidated into a Power BI analytics dashboard.

3. Data Architecture and Schema

The architecture of this project followed a multi-tool data pipeline, where different types of data were processed in the most suitable environment.

3.1 Data Pipeline Architecture

The pipeline consisted of the following components:

Tool	Purpose
PostgreSQL	Store core transactional data
Excel	Manage product master data
Python (Pandas)	Clean unstructured and spatial data
Google Sheets	Store executive target metrics
Power BI	Data integration, modeling, and visualization

3.2 Data Sources: Kaggle – Olist Brazilian E-commerce Dataset

<https://www.kaggle.com/datasets/olistbr/brazilian-ecommerce>

The dataset included the following tables:

- Orders
- Order Items
- Customers
- Sellers
- Products
- Payments
- Reviews
- Geolocation
- Category Name Translation

In Power BI, the dataset was structured using a Star Schema data model to improve analytical performance and simplify relationships between tables.

Fact Table

The central Fact Table (Fact_Sales) was created by integrating data from multiple transactional tables:

- Orders
- Order Items
- Payments
- Reviews

These tables were combined because they represent transactional business events related to each order.

The fact table contains measurable metrics such as:

- Order purchase date
- Freight value

- Payment value
- Delivery dates
- Review scores
- Order item identifiers

These metrics allow analysis of revenue performance, delivery efficiency, and customer satisfaction.

Dimension Tables

Several dimension tables were connected to the fact table using one-to-many relationships.

Dim_Customers

- customer_id
- customer_city
- customer_state
- customer_zip_code_prefix

This table provides customer demographic and geographic information.

Dim_Products

- product_id
- product_category_name_english
- product_weight_g
- product_length_cm
- product_height_cm
- product_width_cm

This table contains product attributes used to analyze product category performance.

Dim_Sellers

- seller_id

- seller_city
- seller_state
- seller_zip_code_prefix

This table enables analysis of seller performance and geographic distribution.

Dim_Geolocation

- geolocation_zip_code_prefix
- geolocation_city

This dimension supports geographic analysis and mapping visualizations.

Dim_Date

A separate date dimension was created using the DAX function: `Dim_Date = CALENDARAUTO()`

This table allows time-based analysis such as:

- Yearly revenue trends
- Monthly sales patterns
- Year-to-date calculations

Dim_Target_Metrics

This table was imported from Google Sheets and contains executive target metrics such as:

- Target Revenue
- Target Fulfillment Rate
- Month-Year

This allows comparison between actual performance and business targets.

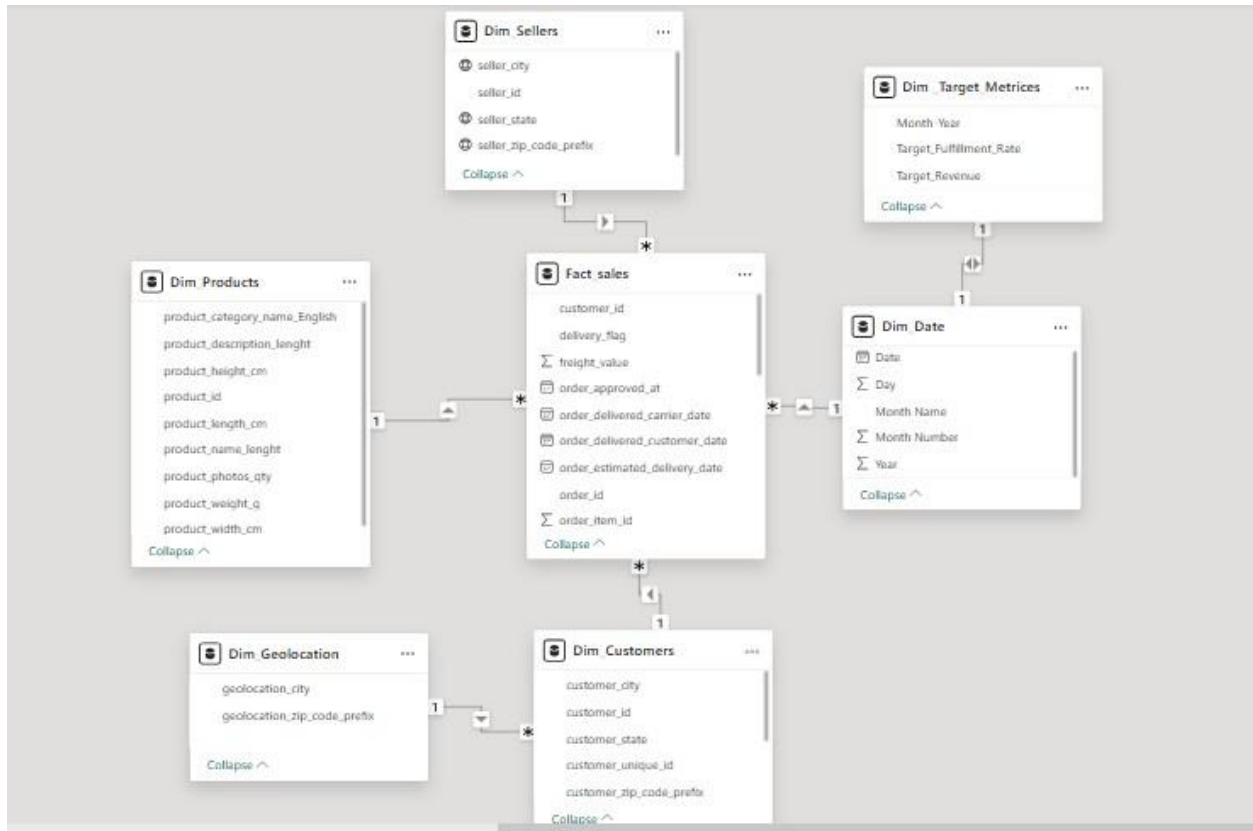


Figure 1: Power BI Star Schema Data Model

4. Data Cleaning and Transformation Methodology

Data cleaning and transformation were performed using Excel, PostgreSQL, Python, and Power BI Power Query.

4.1 Excel Data Cleaning

The Products and Category Translation datasets were cleaned in Excel.

The following functions were used:

- **PROPER()**
Standardized capitalization in product category names.
- **TRIM()**
Removed unnecessary spaces from text fields.

Products with missing physical dimensions (weight, length, width) were identified using filtering and highlighted for review.

4.2 PostgreSQL Data Cleaning

Core transactional tables were stored in PostgreSQL.

Data cleaning operations included:

4.2.1 Geographic Normalization

City names were standardized using SQL functions:

```
UPDATE customers  
SET customer_city = REPLACE(LOWER(customer_city), 'ã', 'a');
```

4.2.2 Handling Missing Values

To prevent calculation errors caused by NULL values, the following function was used:

```
COALESCE(payment_value, 0)
```

This ensured that missing financial values were treated as zero.

4.3 Python Data Cleaning

Python and Pandas were used to process the reviews and geolocation datasets.

4.3.1 Handling Missing Review Comments

Many reviews had missing comment messages. Instead of removing these rows, missing values were replaced using:

```
reviews['review_comment_message'] = reviews['review_comment_message'].fillna('No  
Comment Provided')
```

4.3.2 Removing Geolocation Inconsistencies

The geolocation dataset contained multiple city spellings for the same zip code.

The most frequent city name per zip code was selected using:

```
cleaned_df = df.groupby('geolocation_zip_code_prefix')['geolocation_city'] \
               .agg(pd.Series.mode) \
               .reset_index()
```

This created a cleaner and more consistent geographic dataset.

4.4 Power Query Transformations

Data integration and transformation were also performed in Power BI using Power Query. The English product category translation table was merged with the Products table using a Left Outer Join. Additionally, the order purchase timestamp column was split into Purchase Date and Purchase Time columns to support time-based analysis.

4.5 Data Transformation

Several engineered metrics were created before loading the data into Power BI.

4.5.1 RFM Customer Segmentation (SQL)

Customers were segmented based on:

- **Recency** – How recently the customer made a purchase
- **Frequency** – Number of orders
- **Monetary Value** – Total spending

The NTILE(5) SQL window function was used to assign scores from 1 to 5 for each metric.

This segmentation helps businesses identify:

- Loyal customers

- High-value customers
- Inactive customers

```

• CREATE OR REPLACE VIEW rfm_segmentation AS
• WITH rfm_base AS (
•     SELECT
•         o.customer_id,
•         MAX(o.order_purchase_timestamp::date) AS last_order_date,
•         COUNT(o.order_id) AS frequency,
•         SUM(p.payment_value) AS monetary
•     FROM orders o
•     JOIN payments p
•         ON o.order_id = p.order_id
•     GROUP BY o.customer_id
• )
•
• SELECT
•     customer_id,
•
•     -- Recency (days since last order)
•     CURRENT_DATE - last_order_date AS recency_days,
•
•     -- R score
•     NTILE(5) OVER (ORDER BY CURRENT_DATE - last_order_date DESC) AS r_score,
•
•     -- F score
•     NTILE(5) OVER (ORDER BY frequency ASC) AS f_score,
•
•     -- M score
•     NTILE(5) OVER (ORDER BY monetary ASC) AS m_score
•
• FROM rfm_base;

```

4.5.2 Logistics Delivery Performance (Python)

Delivery performance was analyzed by calculating delivery delays.

Steps included:

1. Converting date columns to datetime format: `pd.to_datetime`
2. Calculating delivery variance: `(orders['order_delivered_customer_date'] - orders['order_estimated_delivery_date']).dt.days`
3. Categorizing delivery status:

- On Time
- Slightly Delayed
- Severely Delayed

This classification allowed analysis of logistics efficiency.

5. Dashboard Visualizations

Two dashboards were created in Power BI following data visualization best practices.

5.1 Page 1: Executive Overview (Z-Pattern Layout)

The first dashboard provides a high-level overview of business performance and is designed primarily for executives and decision-makers.

The layout follows the Z-pattern scanning principle, where users first read information from the top-left to top-right, then diagonally across the page, and finally toward the bottom section.

5.1.1 KPI Indicators

The top section of the dashboard contains key performance indicators that summarize the overall performance of the e-commerce platform.

The KPIs displayed include:

- **Total Revenue:** 16.64M
- **Total Orders:** 99K
- **Total Cost:** 9.99M
- **Profit Margin:** 39.97%
- **Average Order Value (AOV):** 168.69
- **Fulfillment Rate:** 0.98

These metrics allow executives to quickly evaluate the financial health and operational performance of the business.

5.1.2 Revenue Trend Analysis

A line chart was used to visualize Total Revenue and Total Orders over time.

This chart enables decision-makers to:

- Observe sales trends
- Identify peak sales periods
- Monitor fluctuations in revenue

Time-based analysis is important for identifying seasonal patterns and growth trends in e-commerce activity.

To maintain a clear and stable overview, the interaction between this line chart and other visuals (KPI cards and the product category table) has been disabled. This ensures that filtering or selecting values in the chart does not affect the KPI indicators or the category table, allowing executives to always view the overall business summary without accidental filtering.

5.1.3 Product Category Performance

The bottom section of the dashboard contains a table that displays product category performance.

The table includes:

- Product Category Name
- Total Revenue
- Total Orders
- Total Cost

A Top 5 filter based on Total Revenue is applied to highlight the highest-performing product categories.

This visualization helps identify:

- High-performing product categories
- Categories generating the highest revenue
- Categories with lower sales performance

Such insights support inventory management and marketing strategies.

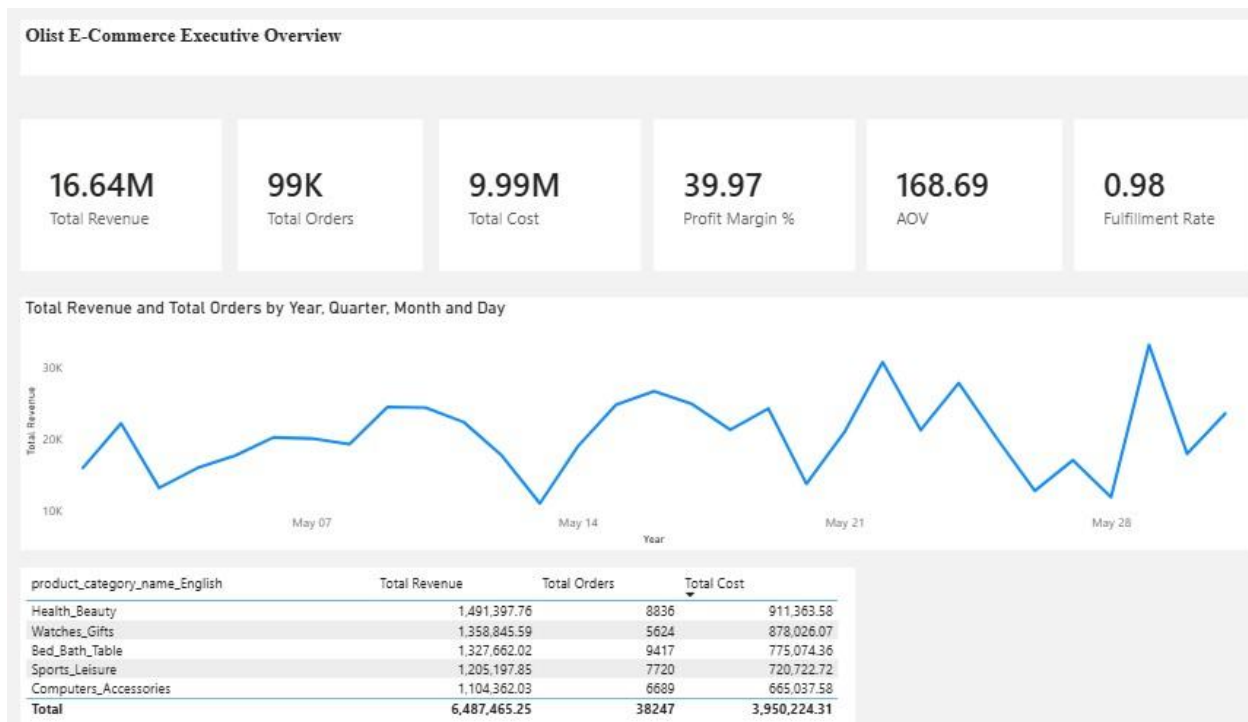


Figure 2: Executive Overview Dashboard (Z-Pattern Layout)

5.2 Page 2: Deep-Dive Analysis (F-Pattern Layout)

The second dashboard was designed to provide a detailed analytical view of the e-commerce data, allowing users to explore performance metrics and identify underlying factors affecting business outcomes.

This dashboard follows the F-pattern layout, which is commonly used for analytical dashboards. In this layout, users first scan the left side of the screen where filtering controls are located, and then move horizontally across the charts and metrics for deeper analysis.

5.2.1 Key Performance Indicators

At the top of the dashboard, several important KPIs are displayed to summarize operational performance.

The key metrics include:

- **Total Revenue** : The total revenue generated from all completed orders in the dataset.
- **Profit Margin (%)**: This metric represents the percentage of profit earned after subtracting operational costs from revenue.
- **Average Delivery Time (Days)**: This KPI measures the average number of days required for orders to be delivered to customers.

These indicators help decision-makers quickly evaluate the financial performance and operational efficiency of the business.

Unlike the first dashboard, these KPI values change dynamically based on the applied filters (slicers). This allows users to analyze performance for specific time periods, locations, product categories, or payment types.

5.2.2 Interactive Filters (Slicers)

To support flexible analysis, several slicers were added on the left side of the dashboard. These slicers allow users to dynamically filter the data and explore specific segments.

The filters include:

- **Order Delivered Date**: Allows users to analyze sales and delivery performance within a selected time range.
- **Customer State**: Filters the data based on the geographic location of customers.
- **Product Category**: Allows analysis of revenue and performance for specific product categories.
- **Payment Type**: Enables users to evaluate how different payment methods influence sales performance.

These interactive filters enable users to perform customized analysis based on their specific business questions.

5.2.3 Decomposition Tree Analysis

A Decomposition Tree visualization was implemented to analyze revenue across multiple hierarchical levels.

The decomposition tree allows users to drill down into the data to understand how revenue is distributed across different dimensions.

The analysis hierarchy includes:

- Customer State
- Customer City
- Product Category

Using this visualization, users can easily identify:

- Which states generate the highest revenue
- Which cities contribute the most to sales
- Which product categories perform best in specific regions

This type of analysis helps businesses detect regional sales trends and product demand patterns.

5.2.4 Analytical Benefits

The Deep-Dive Analysis dashboard allows decision-makers to:

- Investigate revenue patterns across regions
- Analyze product category performance
- Evaluate delivery performance
- Identify profitable market segments
- Understand how payment methods affect sales

By combining interactive filtering with hierarchical drill-down analysis, the dashboard provides a powerful tool for exploring business performance in detail.

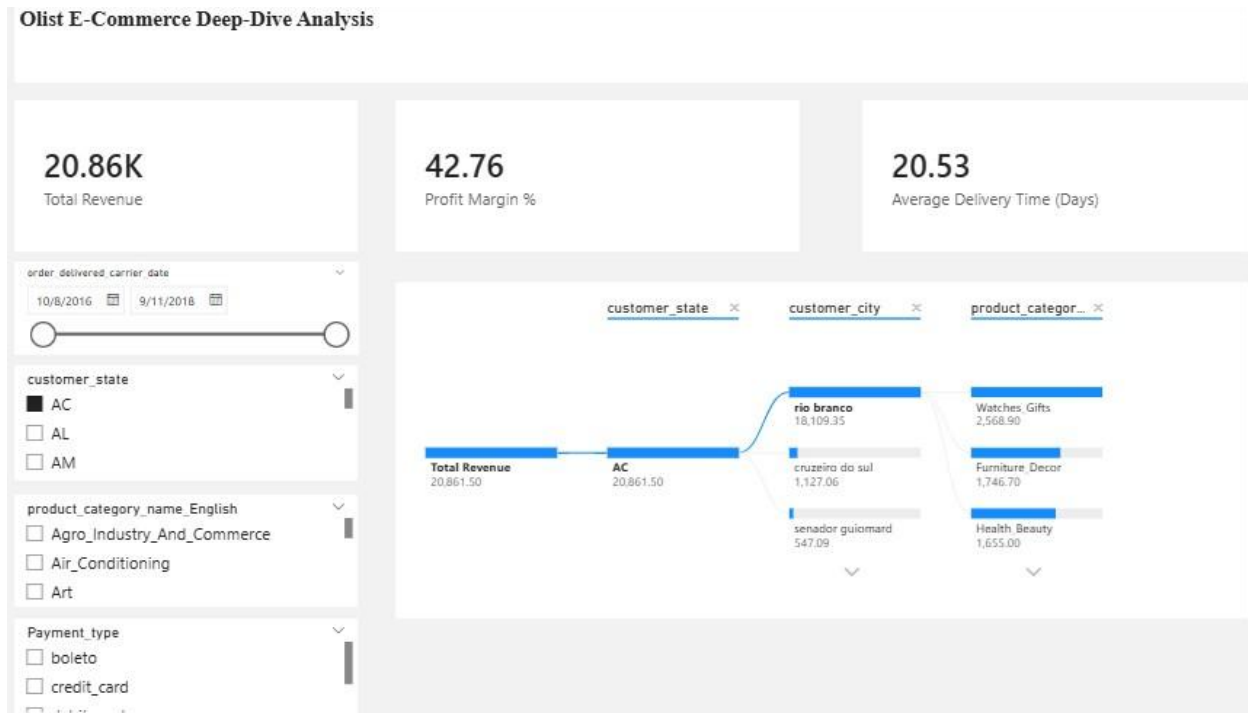


Figure 3: Deep-Dive Analysis Dashboard (F-Pattern Layout)

6. Strategic Recommendations

Based on the analysis performed in this project, several strategic recommendations can be made to improve business performance.

- 1. Improve Logistics in High Delay Regions:** The delivery analysis showed that some regions experience higher delivery delays. The company should investigate logistics partners in those areas and optimize delivery routes or change logistics providers to improve fulfillment performance.
- 2. Strengthen Customer Retention Strategies :** RFM segmentation identified high-value and loyal customers who contribute significantly to revenue. The company should implement loyalty programs, personalized offers, and targeted marketing campaigns to retain these valuable customers.

3. **Focus on High-Performing Product Categories:** The dashboard analysis showed that certain product categories generate higher revenue and order volume. Increasing inventory availability and promotional activities for these high-performing categories can further boost sales.
4. **Optimize Payment Options:** Analysis of payment types can help identify the most preferred payment methods. Businesses should ensure smooth transactions for the most commonly used payment options to improve customer experience.

7. Conclusion

This project successfully demonstrated the development of a complete data analytics pipeline that integrates multiple technologies and transforms raw transactional data into meaningful business insights.

By combining PostgreSQL, Excel, Python, Google Sheets, and Power BI, the project simulated a real enterprise data architecture.

The final Power BI dashboards provide decision makers with interactive tools to monitor revenue performance, customer behavior, and logistics efficiency, enabling better strategic planning and operational improvements.